CANKIRI KARATEKİN UNIVERSITY
FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING

EARLY DIAGNOSIS OF ALZHEIMER'S DISEASE USING DEEP NEURAL NETWORKS

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GRADUATION THESIS REPORT

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I would like to extend my heartfelt thanks to my esteemed advisor, ***Assistant Prof. Zafer CIVELEK***, for his invaluable support, insightful opinions, and suggestions throughout the completion of my graduation project.

ABSTRACT

Alzheimer's disease is a serious neurodegenerative disorder commonly found in the elderly population, leading to progressive deterioration of cognitive functions. The aim of this study is to detect early-stage symptoms of Alzheimer's disease using deep learning techniques such as Convolutional Neural Networks (CNNs) and to assist in expert diagnosis. The performance of various deep learning networks (NasNetLarge, DarkNet-53, VGG19, Inception-V3, GoogleNet, ResNet-101, DenseNet-201, etc.) was compared in this study. The Adam optimization method was applied to each network. This study was conducted using a dataset comprising 896 MildDemented, 2240 VeryMildDemented, 3200 NonDemented, and 64 ModerateDemented labeled images. The images were pre processed through resizing. Simulations conducted in the Matlab environment revealed that the best results were obtained with the DarkNet-53 network architecture optimized with Adam, compared to other networks.

Keywords: Deep learning, Alzheimer's disease, Convolutional neural networks

1. INTRODUCTION

Alzheimer's disease is a neurological disorder characterized by progressive deterioration of cognitive functions and is commonly observed in the elderly population. Currently, early recognition of symptoms and accurate diagnosis of Alzheimer's disease are of paramount importance. However, traditional diagnostic methods are often conducted after symptoms manifest, which may delay the initiation of treatment.

1.1. Objective

The successes achieved in fields such as image processing and data analysis through deep learning methods offer potential tools that can be employed in the diagnosis of Alzheimer's disease. This study aims to evaluate different architectures using deep learning techniques to achieve early diagnosis of Alzheimer's disease.

2. Literature Review

In this study, deep learning tools were developed using Ultra-Performance Liquid Chromatography-Mass Spectrometry (UPLC-MS/MS) based metabolomic data to predict Alzheimer's disease. A total of 177 individuals, including 78 diagnosed with Alzheimer's and 99 normal controls, were selected from the Alzheimer's Disease Neuroimaging Initiative (ADNI) cohort with 150 metabolomic biomarkers. The best deep learning model, consisting of two layers and 18 neurons, predicted Alzheimer's disease with 88.1% accuracy, 89.2% F1 score, and 87.3% AUC **[1]**. Another study investigates the feasibility of shortening the positron emission tomography (PET) scan duration for evaluating amyloid-β levels in Alzheimer's patients. Using a deep learning-based noise removal algorithm called MCDNet-2, developed in this study, superior noise reduction performance was achieved in 5-minute PET images compared to other methods, potentially reducing the PET scan duration from 20 minutes to 5 minutes **[2].**

This article presents a deep learning framework with an embedded feature selection approach for early diagnosis of Alzheimer's disease (AD). Using an AD DNA methylation dataset, four embedded feature selection models are compared to select the most relevant features. An Enhanced Deep Recurrent Neural Network (EDRNN) is implemented and compared with existing classification models, showing a significant improvement in classification accuracy **[3]**. This study utilizes a multi-modal deep learning library to predict the progression of Alzheimer's disease based on time series data. The library combines a 3D convolutional neural network (CNN) with bidirectional recurrent neural networks (BRNN) to capture intra-slice and inter-slice patterns in 3D MR images. The study also explores fusion of MRI data with cross-sectional biomarkers and introduces an interpretability approach to enhance model interpretability. Experimental results demonstrate high accuracy, sensitivity, recall, and area under the receiver operating characteristic curve for the proposed library **[22].** Additionally, there is a study presenting a deep learning model named C3d-LSTM for detecting Alzheimer's disease based on 4D functional magnetic resonance imaging (fMRI) data. The model combines 3D convolutional neural networks (CNNs) to extract spatial features from each volume of fMRI image sequences and a long short-term memory (LSTM) network to capture temporal information in the data. The results show that directly using 4D fMRI data with the proposed method improves Alzheimer's disease detection compared to methods using functional connectivity or 2D/3D fMRI data **[16]**. Moreover, this study proposes a multi-task deep learning model based on time series data to detect the progression of Alzheimer's disease. The model utilizes stacked convolutional neural networks (CNNs) and bidirectional long short-term memory (BiLSTM) networks to extract features from different data modalities and predicts multiple variables related to the progression of Alzheimer's disease and cognitive scores regression tasks **[17].**

Furthermore, a study discusses a deep learning and fuzzy logic-based classification strategy for early diagnosis of Alzheimer's disease in individuals with mild cognitive impairment (MCI) using structural MRI features. The study compares the accuracy of deep learning strategies with other machine learning approaches and ranks third in an international competition for predicting MCI from MRI data **[8].** This article introduces a new deep learning method called Fuzzy-VGG for predicting the staging of Alzheimer's disease based on structural brain magnetic resonance imaging (sMRI) data. The method utilizes blurring theory to rearrange image pixels according to grayscale levels and emphasize the most important local information. Subsequently, a VGG-based primary framework is used to train datasets, and a two-stage cropping strategy is employed to enhance datasets and improve training results. The results demonstrate improved classification performance and accelerated model convergence **[4].** Furthermore, there is a study discussing the use of a deep Siamese convolutional neural network (CNN) for multi-class classification of Alzheimer's disease. Researchers developed a model inspired by VGG-16 to classify dementia stages using magnetic resonance imaging (MRI) data. They addressed the problem of limited image samples in datasets by using augmentation approaches. The proposed model achieved a perfect test accuracy of 99.05% and outperformed state-of-the-art models in terms of performance, efficiency, and accuracy **[32].** Finally, this article presents a deep learning model named VGG-TSwinformer for early prediction of Alzheimer's disease (AD) using structural brain magnetic resonance imaging (sMRI) data. The model combines convolutional neural networks (CNNs) and Transformer to analyze longitudinal sMRI images and accurately predict the progression trend of mild cognitive impairment (MCI), which is a transitional state between normal aging and AD. The proposed model achieves better results compared to other cross-sectional studies, demonstrating the potential to enhance the diagnostic efficiency of AD **[12].**

This study presents a multimodal deep learning model for assisting the diagnosis of Alzheimer's disease. Using the deep learning model, it demonstrates that assisted diagnosis with multimodal neuroimaging, when combined with clinical neuropsychological diagnosis, enhances accuracy. Experiments show that the proposed method is effective on the Alzheimer's Disease Neuroimaging Initiative (ADNI) database and can achieve excellent diagnostic efficiency [5]. Additionally, this study introduces a patch-based deep multimodal learning (PDMML) framework for diagnosing Alzheimer's disease using multi-view neuroimaging. The framework integrates patch-level multimodal imaging features to capture brain disease representations. A discriminative location discovery strategy is employed to filter normal regions without prior knowledge. The proposed method shows promising results in discriminative location discovery and brain disease diagnosis **[26].**

A unique color-coded visualization system called ML4VisAD, combining machine learning and deep learning techniques, is introduced to predict the progression of Alzheimer's disease in a longitudinal study over 2 years. The system utilizes fundamental measurements and multimodal inputs such as neuroimaging data, neuropsychological test scores, and biomarkers to generate color-coded visual images reflecting disease progression at different time points. The study demonstrates the effectiveness of the system in accurately diagnosing and predicting Alzheimer's disease and highlights the challenges of multi-class classification and regression analysis **[6]**. This study proposes a new computer-aided diagnosis (CAD) system for Alzheimer's disease using magnetic resonance imaging (MRI) data. The proposed system includes a segmentation method that addresses temporal changes in intensity homogeneities, a feature extraction approach combining texture, edge, color, and intensity features, and a hybrid deep learning model for classification. Simulation results indicate that the proposed CAD system outperforms existing systems in terms of accuracy, sensitivity, and specificity **[24].**

This study proposes a deep learning model called FDN-ADNet for early diagnosis of Alzheimer's disease using MRI scans. The model utilizes sagittal plane slices from 3D MRI images for feature extraction and then applies a fuzzy hyperplane-based twin support vector machine (FLS-TWSVM) for classification. The model achieves high accuracy compared to state-of-the-art networks for different classification tasks **[7].** This study discusses the use of machine learning and deep learning algorithms for the diagnosis of Alzheimer's disease. It reviews previous research in this field and compares the effectiveness and error rates of various algorithms. The aim is to provide researchers and clinicians with the necessary machine learning models for classifying Alzheimer's disease. The article also emphasizes the importance of using multiple features from different MRI analysis techniques to improve classification accuracy. The dataset used in the study includes longitudinal PET scans and MRI techniques from the Alzheimer's Disease Neuroimaging Initiative (ADNI). The study reviews the literature on different machine learning techniques used for Alzheimer's diagnosis, including support vector machines (SVM), artificial neural networks (ANN), and deep learning. It also discusses the importance of image modality and kernel functions in MRI image classification. The article concludes by suggesting that SVM is a highly stable and widely used method for Alzheimer's diagnosis and that combining multiple features can improve classification accuracy **[30]**.

This study investigates the use of deep learning techniques, particularly deep residual neural networks (ResNet), for analyzing electroencephalogram (EEG) signals to predict the progression of Alzheimer's disease. ResNet models successfully capture nonlinear features and achieve high accuracy in predicting the progression of Alzheimer's disease. The models also exhibit similar peak activations to previous Alzheimer's disease reports and outperform existing methods such as support vector machines and stacked autoencoders **[9].** This study investigates the use of deep learning for predicting Alzheimer's disease using electronic medical record (EMR) data. The study demonstrates that the application of medical domain knowledge in data preprocessing and positive dataset selection significantly improves the prediction of Alzheimer's disease compared to naive models **[10].**

Using Micro-Optical Sectioning Tomography (MOST) images, this study presents a deep learning-based method for detecting and counting neuronal somas in Alzheimer's disease mouse brains. The proposed method achieves faster and more accurate results compared to manual methods and existing software. The study also reveals a decrease in neuron counts in specific brain regions in Alzheimer's diseased mice consistent with previous experimental findings. This method provides a potential tool for analyzing neuron changes during the progression of Alzheimer's disease **[11].** This study introduces a deep learning-based model for classifying healthy aging controls, mild cognitive impairment (MCI), and Alzheimer's disease using a fusion of magnetic resonance imaging (MRI) and positron emission tomography (PET) scans. The model demonstrates potential for automated diagnosis of dementia stages by achieving high accuracy in distinguishing between these groups **[25].** This study discusses advances in the deep neuropathological phenotyping of Alzheimer's disease, including the history of Alzheimer's disease neuropathology, microscopy and staining techniques, in vivo biomarkers, and machine learning algorithms. It highlights the importance of precision medicine approaches and potential new treatments for AD. Furthermore, it emphasizes the role of whole slide imaging (WSI) in providing deeper phenotyping of Alzheimer's neuropathology and discusses other scientific advances such as proteomic, PET imaging biomarkers, digital spatial profiling, and machine learning algorithms to enhance Alzheimer's research **[31].**

This study discusses a deep learning library for improving the diagnosis of Alzheimer's disease by incorporating neuropsychiatric symptoms (NPS). The library utilizes multimodal inputs such as structural MRI, behavioral scores, age, and gender information. The model achieves high accuracy in classifying Alzheimer's disease and cognitively normal individuals **[13].** This study discusses the use of deep learning techniques for early diagnosis of Alzheimer's disease based on electroencephalogram (EEG) spectral images. The authors propose a multitask learning strategy based on discriminative convolutional higher-order Boltzmann Machines with hybrid feature maps. They demonstrate that this strategy can extract more abstract features from limited EEG spectral images and achieve better generalization and improved results compared to existing methods **[14].** This study presents a deep learning approach for diagnosing Alzheimer's disease using electroencephalogram (EEG) signals. The authors propose a framework comprising a Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) generative model to predict latent factors of brain activity from multi-channel EEG data, along with a Spatio-Temporal Autoencoder (STAE). The study shows that the proposed deep learning-based diagnostic scheme achieves high accuracy, sensitivity, and specificity in classifying Alzheimer's patients and normal subjects. The STAE model also outperforms other approaches in predicting hidden brain dynamics **[27].**This study introduces a hybrid model for diagnosing Alzheimer's disease using electroencephalography (EEG) signals. The model combines deep ensemble learning (DEL) and 2D convolutional neural networks (2D-CNN) to accurately classify EEG signals of Alzheimer's disease and healthy control subjects. The proposed DEL model achieves high accuracy in

Alzheimer's disease classification, reaching an average of 97.9% accuracy in cross-validation **[28].**

Using Magnetic Resonance Imaging (MRI) data, this study discusses the use of unsupervised deep learning technology for automatic prediction of Alzheimer's Disease. The authors propose a method using unsupervised Convolutional Neural Networks (CNNs) for feature extraction and an unsupervised predictor for final diagnosis. Experimental results demonstrate prediction outcomes for MCI against Alzheimer's Disease and MCI against NC using both single-slice and Three Orthogonal Panel (TOP) data **[15].** This study discusses a deep learning convolution technique for recognizing Alzheimer's disease based on magnetic resonance imaging (MRI) and functional MRI (fMRI) data. This technique uses convolutional neural networks to extract low-level, invariant, and undeniable features from images. The authors demonstrate the effectiveness of their approach in accurately classifying Alzheimer's disease from normal control data **[29].** This study encompasses the diagnosis process of Alzheimer's disease using deep learning techniques applied to the analysis of magnetic resonance imaging (MRI) scans, while the monitoring stage employs inertial sensors attached to the body to monitor patients' daily life activities. The results show a significant improvement in the accuracy of Alzheimer's disease diagnosis and high accuracy in its classification **[18].**

Using Magnetic Resonance Imaging (MRI), this study integrates voxel-based, region-based, and patch-based approaches in a unified framework for diagnosing Alzheimer's disease or mild cognitive impairment (MCI). By partitioning the brain into predefined regions and inferring complex relationships between voxels in each region, a "regional abnormality representation" is obtained. This representation allows the interpretation and visualization of symptomatic observations of subjects with AD or MCI. The method achieves state-of-the-art performance in classification tasks using the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset **[19].**

This study utilizes a deep learning library for classifying Alzheimer's disease. The library includes feature learning, ranking of basic classifiers, and optimization of layers to improve classification accuracy. In experiments conducted with clinical datasets, the proposed library outperforms six well-known ensemble approaches **[20].** Mobility data and deep learning models are employed in this research to determine the stage of Alzheimer's Disease patients. Data from 35 Alzheimer's Disease patients using smartphones were collected, and the data were labeled with the disease stage. A Convolutional Neural Network (CNN) model was used to recognize patterns in accelerometer changes recorded during daily activities. The CNN-based method achieved a 90.91% accuracy rate and improved upon results obtained by traditional feature-based classifiers. The study demonstrates the potential value of mobility data for the treatment and monitoring of Alzheimer's Disease patients **[21].** This study explores the use of optimization techniques and deep learning methods to analyze brain subregions in Alzheimer's Disease. The aim is to develop an automated process for identifying Alzheimer's Disease by segmenting regions such as White Matter, Corpus Callosum, Gray Matter, and Hippocampus. Four optimization algorithms (Genetic Algorithm, Particle Swarm Optimization, Gray Wolf Optimization, and Cuckoo Search) are compared, with Gray Wolf Optimization showing good results in accurately segmenting brain subregions. The segmented regions are then classified using a deep learning classifier, demonstrating the highest classification accuracy for the Hippocampus region **[23].** Various clinical data are used in this study to highlight a deep learning library for the diagnosis of Alzheimer's Disease and dementia. The developed models have shown comparable accuracy to neurologists and neuroradiologists. Disease-specific models in brain degeneration are identified using interpretable methods in computer vision. This research demonstrates the potential of deep learning in improving cognitive impairment diagnosis **[33].**

This study compares different representations of the hippocampus to analyze Alzheimer's Disease. Mesh-based representations are preferred for ease of interpretability, while Region of Interest (ROI) augmentation performs well. The study emphasizes the importance of representation selection in terms of prediction performance and interpretability and recommends mesh-based representations for their benefits **[34].** The study discusses the combined use of deep learning techniques with neuroimaging and genomic data for the diagnosis and prediction of Alzheimer's Disease. The authors highlight various studies that use deep learning algorithms to predict Alzheimer's Disease using either genomic or neuroimaging data. They also outline integrative studies that combine both neuroimaging and genomic data **[35].**

3. Materials and Methods

3.1. *Dataset*

In this thesis, a dataset downloaded from the Kaggle website was utilized. The dataset consists of 896 MildDemented, 2240 VeryMildDemented, 3200 Non Demented, and 64 ModerateDemented images. The link to the dataset is: https://www.kaggle.com/datasets/tourist55/alzheimers-dataset-4-class-of-images

3.2. *Preprocessing*

MRG segmentation images, varying in size, were resized to dimensions of 100x100x1. Subsequently, they were subjected to random rotation between -45 degrees to 45 degrees. Afterward, some images were randomly resized between 0.5 to 1.

3.3. *Transfer Learning*

In transfer learning, 13 different pre-trained networks were used. The pre-trained weights of these networks were directly adopted, and no changes were made to the convolutional layers. Only adjustments were made to the fully connected layers to match the dataset and the number of classes. The simulation, conducted in the Matlab environment, was configured with mini-batch size=10, max epoch=8, learning rate=0.0001, and validation frequency=5.

3.3.1. NasNetLarge

NasNetLarge, developed by the Google Brain team, is designed to achieve high accuracy on large datasets. The architecture was initially trained on the CIFAR-10 dataset, and the best convolutional layers were identified and then applied to the ImageNet dataset. Neural Architecture Search (NAS) method was used to automatically design the most suitable architectures for specific tasks. NAS consists of three main components: search space, search strategy, and evaluation strategy. The search space defines potential model structures and was created with a cell-based approach for NasNetLarge. Reinforcement learning was used as the search strategy, where an RNN agent discovers cell architectures and optimizes them based on a reward function. The evaluation strategy determines the performance of these cells and selects the best structures. NasNetLarge significantly improved generalization with a regularization technique called ScheduledDropPath, which randomly disables some paths during training to prevent overfitting. Additionally, compared to streamlined architectures targeting mobile and embedded platforms, it outperformed them with lower computational budgets. One significant advantage of this architecture is the automatic weight adjustment on large datasets without human intervention. NasNetLarge outperformed many human-designed models in both accuracy and efficiency, demonstrating the power and potential of the NAS method. Paper

3.3.2. NasNetMobile

NasNetMobile is a smaller and lighter model of the NasNet family. It adopts the same principles as NasNetLarge and is designed using Neural Architecture Search (NAS). It has lower parameter coefficients and memory usage.

3.3.3. VGG19

VGG-19, developed by a group of researchers from the University of Oxford, has the same basic architecture as the VGG16 model except for an additional 3 convolutional layers. VGG19 achieves significantly better performance than the AlexNet by using several 3x3 filter size convolutional layers instead of larger filter sizes. The image passes through a series of convolutional layers using 3x3-pixel small receptive field filters. The convolution step is fixed at 1 pixel, and spatial resolution is preserved using 1-pixel padding. Max pooling is performed with a 2x2-pixel window and a stride of 2.

The convolutional layers in VGG19 are followed by 3 Fully Connected (FC) layers: the first two have 4096 channels, and the third layer performs a 1000-class ILSVRC classification. The final layer is a softmax layer. All hidden layers use the ReLU non-linear activation function. Except for one, none of the networks contain Local Response Normalization (LRN) because it does not enhance performance and increases memory consumption. An important advantage of VGG19 is using smaller filters by increasing depth, thereby achieving better performance. Paper.

3.3.4. GoogleNet

GoogleNet is a deep learning architecture developed by Google that won first place in the 2014 ImageNet competition. Officially named Inception-V1, the main components of this architecture are the Inception modules, which apply convolutions of different sizes simultaneously to create richer feature maps. These modules combine 1x1, 3x3, and 5x5 convolution layers to capture both local and global features. Additionally, 1x1 convolutions are used throughout the network to reduce dimensionality, thereby lowering computational costs and making the model more efficient. GoogleNet consists of a total of 22 layers, including operations such as max-pooling and average-pooling between these layers. These features summarize and reduce dimensions, enhancing the model's generalization ability. As a result, despite having a deep and wide structure, GoogleNet offers a lightweight and fast model by optimizing the number of parameters. Paper.

3.3.5. Inception-V3

Inception-V3, developed by Google, is a deep learning architecture designed to achieve high accuracy in the ImageNet competition. Inception-V3 uses an advanced version of other Inception modules, featuring factorized convolutions, auxiliary classifiers to facilitate gradient propagation, and strategies to reduce grid sizes for more efficient parameter usage. Factorized convolutions involve dividing larger filters into smaller ones to reduce the computational load. For example, a 5x5 convolution can be split into two 3x3 convolutions. Additionally, auxiliary classifiers are used in intermediate layers to improve gradient flow throughout the network, facilitating the training of deep networks. Inception-V3 was trained using the RMSProp optimization algorithm, and techniques such as data normalization and resizing were used in image preprocessing to enhance accuracy. Paper.

3.3.6. ResNet-101

ResNet-101, developed by Microsoft as part of the ResNet (Residual Network) architecture, is a 101-layer deep network that uses residual connections. These connections allow the output of each layer to be added to the output of several layers later, aiding in better gradient propagation and enabling deeper networks. ResNet-101 primarily consists of a series of residual blocks, each containing 1x1, 3x3, and again 1x1 convolution layers. This structure increases computational efficiency and enables the learning of finer details due to its depth. In a residual block, the input x is added directly to the output ($y=F(x,\{Wi\})+x$), minimizing information loss. This way of operation of residual blocks provides an effective solution to the problem of "vanishing gradient" seen in deep networks, where gradients approach zero as the network deepens, leading to learning cessation. Paper.

3.3.7. DenseNet-201

This architecture, with 201 layers, utilizes dense connections, allowing each layer to access feature maps from all previous layers. This approach improves information flow, facilitates more efficient gradient propagation, and thus makes training deep networks easier. Another advantage of dense connections is their ability to achieve high performance even with smaller and limited datasets. DenseNet201 consists of a series of "Dense Block" and "Transition Layer" layers. Dense Blocks concatenate the outputs of all previous layers, while Transition Layers perform dimensionality reduction operations. This makes the network more compact and efficient in terms of computation, allowing feature reuse. Additionally, the DenseNet architecture prevents overfitting with regularization techniques such as data augmentation and dropout. Paper.

3.3.8. EfficientNet-B0

EfficientNet-B0 is a deep learning architecture developed by the Google Brain team, optimizing model scaling. EfficientNet uses a method called Compound Scaling, which balances the scaling of width, depth, and resolution dimensions used in model scaling. This method optimizes the performance and efficiency of the model. The foundation of EfficientNet-B0 comprises MBConv blocks, which combine deep separable convolutions and squeeze-and-excitation mechanisms to reduce the number of parameters while increasing computational efficiency. The Swish activation function also facilitates smoother gradient flows, aiding in faster and better model training. Paper.

3.3.9. Xception

Xception, developed by Google, is a deep learning architecture that stands for "Extreme Inception." Xception splits the standard convolution operation into depthwise separable convolutions and pointwise convolutions. In the first stage, deep separable convolutions are applied to extract features independently from each channel. In the second stage, pointwise convolutions are used to combine these features. This method reduces the number of parameters and computational costs while improving the model's efficiency. Xception consists of 36 deep separable convolution layers, divided into 14 modules. Each module contains direct connections between input and output, facilitating more efficient gradient propagation and enabling deeper networks. Paper.

3.3.10. MobileNetV2

MobileNetV2 is a fast deep learning architecture optimized for mobile devices, developed by Google. Based on an inverted residual structure where residual connections are between bottleneck layers, MobileNetV2's architecture comprises a 32-filter initial full convolutional layer followed by 19 residual bottleneck layers. Linear Bottlenecks minimize information loss by applying a linear transformation at the input and output of each block. Shortcut connections enable more efficient gradient propagation. The model provides high efficiency and accuracy in real-time applications. Paper.

3.3.11. Inception-ResNet-V2

Inception-ResNet-V2, developed by Google, is a deep learning model that combines the Inception and Residual Network (ResNet) architectures. This architecture consists of a total of 164 layers. It combines the flexibility of Inception modules and the feature of residual connections that facilitate training deep networks. Inception blocks use 1x1 convolutions to reduce multiple filters and kernel sizes. Residual connections provide a solution to the vanishing gradient problem encountered in training deep networks by directly adding the input to the output, enabling more efficient learning of deep layers. The model bears similarities to Inception-V4 but improves performance by adding residual connections, achieving higher accuracy with less computational cost. Paper.

3.3.12. DarkNet-19

DarkNet19 is a lightweight and fast deep learning architecture that forms the basis of the You Only Look Once (YOLO) algorithm's first version. Comprising 19 layers, DarkNet19 consists of 3x3 convolution layers and maximum pooling layers. Batch Normalization and Leaky ReLU activation functions are used in each convolution layer to optimize the accuracy and training time of the model. Due to its relatively fewer layers, it trains faster and requires less computational power. Paper.

3.3.13. *DarkNet53*

DarkNet53 Darknet-53 is a convolutional neural network architecture that forms the basis of the YOLOv3 object detection model. Developed in 2018, this architecture consists of 53 convolution layers. Compared to previous DarkNet versions, this architecture uses more complex structures and adds residual connections. In DarkNet-53, feature extraction is performed with 3x3 and 1x1 convolution layers, and each layer uses Batch Normalization and Leaky ReLU activation function. The model facilitates the training of deeper layers and ensures more efficient gradient propagation through residual connections. Paper

3.4. *Eniyileme Yöntemi - Adam*

Optimization Method - Adam Adam (Adaptive Moment Estimation) is an optimization method that updates model parameters with adaptive learning rates using both first and second moment estimates (mean and squared mean). Due to these features, Adam ensures rapid and efficient convergence and performs well across a wide range of applications. Link.

The Initial Gradient is calculated as:

$$g_t = \nabla_\theta f_t\left(\theta_{t-1}\right)$$

The First Moment Estimate (Momentum) is:

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t)$$

The Second Moment Estimate (Squared Gradient) is:

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2)$$

Bias Corrections are applied as:

$$\widehat{m}_t = \frac{m_t}{1-B_1^t} \qquad \hat{v}_t = \frac{v_t}{1-B_2^t}$$

Parameter updates are computed as:

$$\theta_{t+1} = \theta_t - \frac{\alpha \cdot \widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}$$

Initially, $m_0 = 0$ and $v_0 = 0$ are initialized. In this case, the values of m_t and v_t will be low in the first iterations and will not accurately represent the moment estimates. Bias correction ensures that as t increases and $1 - B_1^t$ and $1 - B_2^t$ approach 1, the values of m_t and v_t are more accurately estimated.

- t is the iteration number,
- m_t is the first moment estimate at time step t

- v_t is the second moment estimate at time step t,
- β_1 , is the decay rate of m_t (typically set to approximately 0.9),
- β_2 , is the decay rate of v_t (typically set to approximately 0.999)
- θ_t represents the parameters of the model,
- g_t is the gradient of the loss function,
- α is the learning rate,
- ϵ represents a small number to prevent division by zero (typically set to 10^−8).

3.5. *Evaluation Metrics*

The prediction performances of three commonly used classification methods, namely Accuracy (ACC), Precision (PR), and Sensitivity (SN), have been evaluated in this study. The equations for these metrics are listed below:

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

$$PR = \frac{TP}{TP+FP} \tag{2}$$

$$SN = \frac{TP}{TP+FN} \tag{3}$$

In Equations (1), (2), and (3), the abbreviations TP: True Positive, TN: True Negative, FP: False Positive, FN: False Negative are used.

4. Results

In this study, the effectiveness of deep learning techniques for the early diagnosis of Alzheimer's disease has been investigated. Experiments conducted using various deep neural network architectures revealed that the DarkNet-53 architecture outperforms other networks. Particularly, the DarkNet-53 network achieved the highest values in test accuracy, precision, and sensitivity metrics. The results can be examined in Table 1.

Table 1: Classification Results of Existing Architectures

Network	Solver	Training Accuracy	Validation Rate	Test Accuracy	Test Precision	Test Sensitivity
NasNetLarge	adam	80	97.93	0,9793	0,9852	0,9777
DarkNet-53	adam	90	**99.29**	**0,9930**	**0,9936**	**0,9933**
VGG19	adam	50	36.36	0,3636	0,0909	0,2500
Inception-V3	adam	90	68.81	0,7113	0,7366	0,6960
GoogleNet	adam	70	66.92	0,6626	0,8368	0,6294
ResNet-101	adam	100	98.27	0,9826	0,9848	0,9814
DenseNet-201	adam	90	88.61	0,9166	0,9197	0,9188
EfficientNet-B0	adam	90	98.11	0,9810	0,9771	0,9834
Xception	adam	100	94.80	0,9486	0,9556	0,9564
NasNetMobile	adam	100	97.86	0,9785	0,9769	0,9806
DarkNet19	adam	100	98.55	0,9855	0,9877	0,9827
MobileNetV2	adam	90	98.57	0,9857	0,9855	0,9864
Inception-ResNet-V2	adam	100	99.16	0,9916	0,9911	0,9926

The findings obtained from the experiments can be summarized as follows:

1. **DarkNet-53 Network:** It has been the most successful model with the highest test accuracy (99.30%), precision (99.36%), and sensitivity (99.33%) values.
2. **Inception-ResNet-V2** and **MobileNetV2**: These models have demonstrated the highest performance after DarkNet-53.
3. **VGG19** and **GoogleNet:** These models have shown lower performance compared to others.

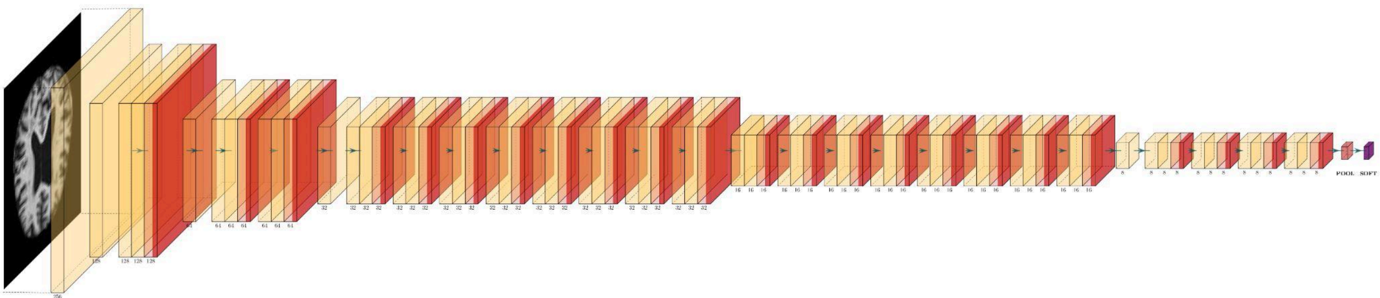

Figure 1: DarkNet-53 Architecture

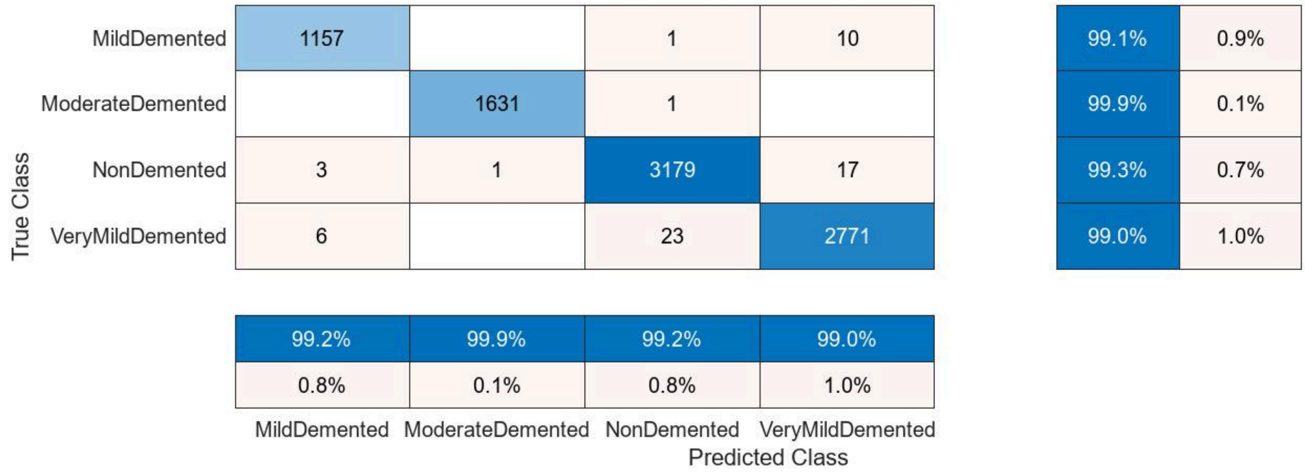

Figure 2: DarkNet-53 results confusion matrix

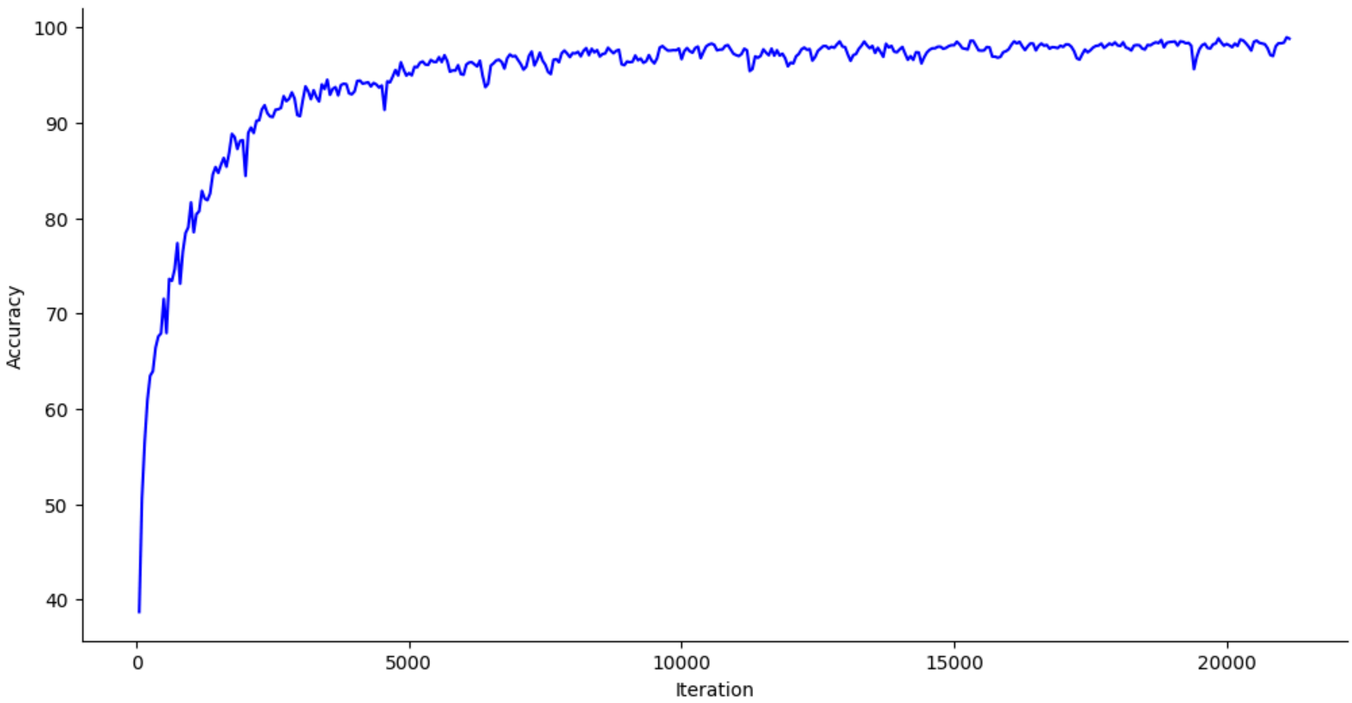

Figure 4: Progression of DarkNet-53 training accuracy value

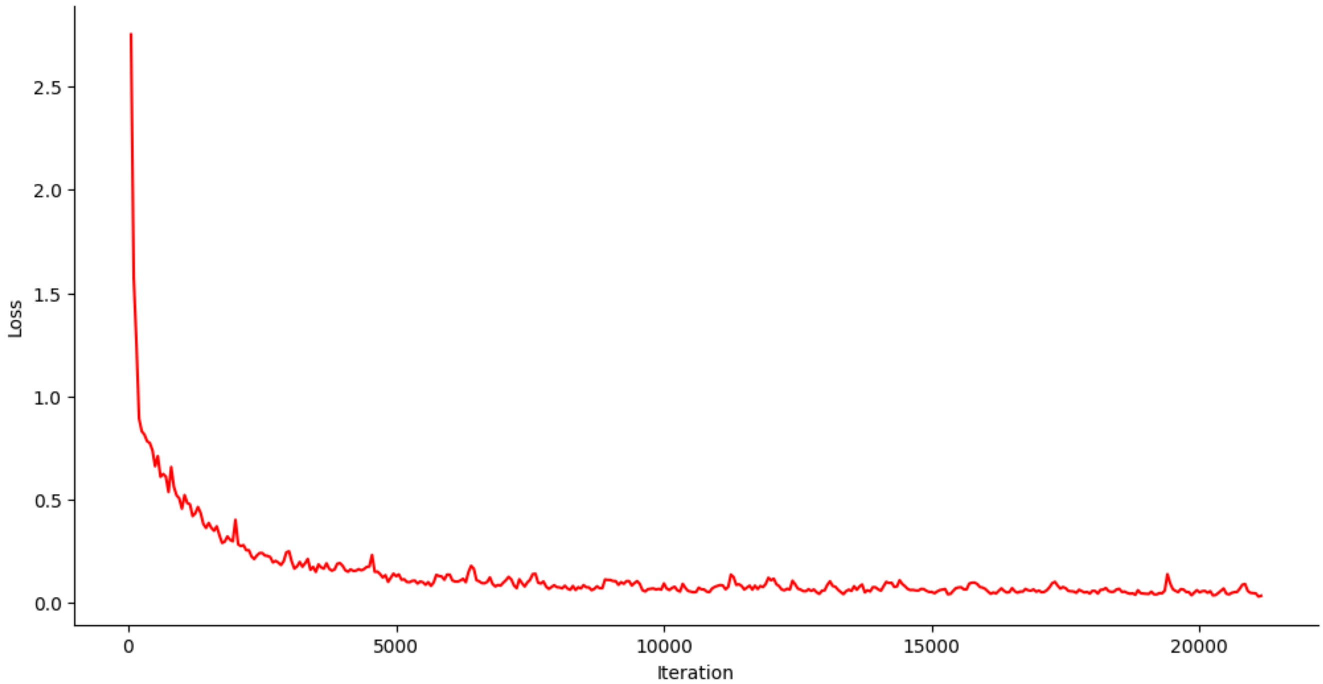

Figure 5: Progression of DarkNet-53 training loss function

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